Yatharth

Technical Design Document: lab report delivery portal for Company A, Vijayawada

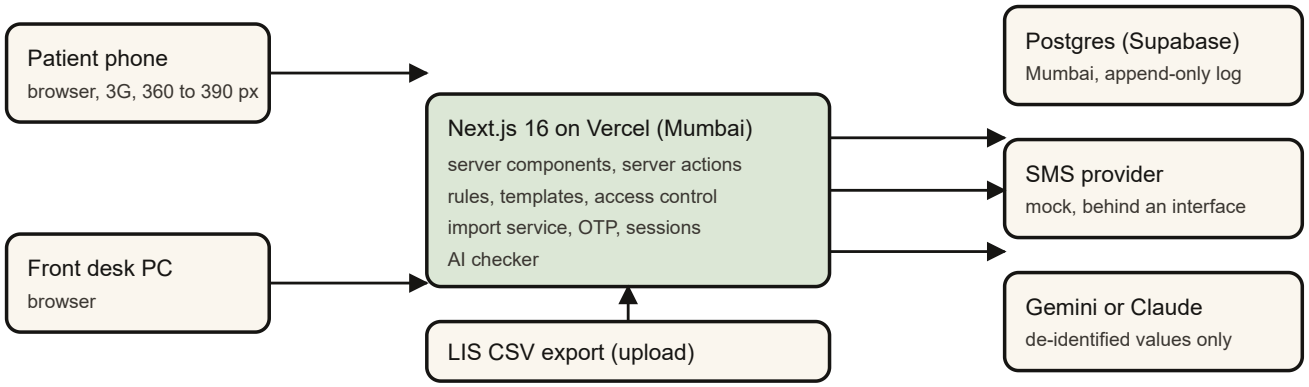
GitHub: https://github.com/shirii8/yatharth_Labs
Live app: https://yatharth-labs-beryl.vercel.app
Demo logins: in deployment.md, section 3.
Built by shirii8, 4 to 9 November 2026. All patient data is synthetic. Version a2b3c63.

1. What was built

The core slice from the brief works end to end: an LIS CSV import, an automatic report link on sign-off, a patient portal with mobile number and OTP, a report view with a plain-language summary of out-of-range values, and a front-desk console that finds a patient, resends a link, shows delivery status and the access log. Roles: patient, front-desk staff, a lab admin who approves wording, reviews the AI and reads the delivery report, and a referring doctor. Beyond the core slice, both optional must-haves are built too: the referring doctor view and the delivery and access report.

Brief, must have	Status
Patient login by mobile and OTP, orders with status, PDF download	Built. OTP with lockout, family on one number, status per report, PDF (a marked stand-in, A7).
Automatic SMS with an expiring link when a report is signed, fed by the LIS export	Built, with SMS mocked. Link created in the same transaction as sign-off; 72 hours; one report per link.
Plain-language summary of out-of-range values in English, Telugu and Hindi	Built. Rules plus locked templates; AI paragraph for tests without locked wording, checked before showing. Telugu approved from the design copy, Hindi waiting for approval to show the approval step. The demo catalogue has 4 tests, 2 of them locked; the brief's 30 locked tests need the lab's own list and doctor-approved wording (section 11).
Front-desk console: find a patient, resend links, delivery status, access log	Built, plus callbacks, flags, add a family member, wording approvals and the AI review.
Referring doctor view of the last 90 days	Built (optional depth). A doctor role tied to one referring doctor: their patients only, last 90 days, full values including critical ones, the signed PDF, and whether the lab has reached a critical patient yet. Every opening is logged.
Delivery and access report for the operations head	Built (optional depth). 7, 30 or 90 days against the brief's targets: share opened online vs 80%, time to first open, critical calls, wrong-person signals, the funnel from signing to reading, and who opened what.

2. Architecture and why



One Next.js app, server rendered. Patient pages are rendered on the server and ship little JavaScript, which suits old phones on 3G. UI and server logic share one language (TypeScript), which is the fastest way for one person to deliver a tested slice in a week. Django or Laravel with Postgres would also fit; Next.js was chosen for server rendering, one language end to end, and speed for a solo build.

Next.js 16, not 14. The plan said 14, but `npm audit` showed critical advisories on the 14 line with no fix. 16.4.5 has a clean production audit.

Structure. `lib/domain` holds pure rules (flags, critical limits, tokens, phone, OTP), each tested on its own. `lib/import`, `lib/auth`, `lib/summary`, `lib/ai`, `lib/patient`, `lib/staff` hold services that use the database. `app/` routes are thin. SMS and AI sit behind small interfaces (`SmsProvider`, `SummaryWriter`), so a real SMS gateway or an India-hosted model is one more class.

Hosting in India. The database is in Supabase's Mumbai region and Vercel functions are pinned to Mumbai (`bom1`). The AI receives no personal data (section 5).

3. Data model

17 tables in Prisma 7 (one schema file per group, one commit per group), changed only by forward migrations. Highlights:

Table	Holds	Design choice
Household	One mobile number	Stored as an HMAC for lookup, AES-256-GCM ciphertext for sending SMS, and a masked form for display. Consent version and time; when first-time setup was finished.
Patient	A family member	Unique by household and name: one number covers a family. The relation to the number holder comes from the LIS and can be corrected by the number holder from a fixed list (A29); re-import never overwrites it.
StaffUser, ReferringDoctor	Who signs in with a password	Roles FRONT_DESK, LAB_ADMIN and DOCTOR. A doctor account points at exactly one ReferringDoctor (unique), which is how every doctor query is scoped.
TestDefinition, Analyte	Tests and what they measure	Critical limits on the analyte; a locked flag for tests with approved wording.

Table	Holds	Design choice
Report, Result	One test on one order	Order plus test is unique (re-import is safe). Each result keeps the range printed on its report and the value as printed; the flag is computed by rule on write.
ReportLink	The SMS link	Only the token's hash is stored. Delivery, opening and revoke times.
OtpChallenge	One code	Hash only, attempts, lock. Not linked to a household, so unknown numbers look the same.
Session	A signed-in browser	Server-side, so it can be ended at once. Kinds: PATIENT, LINK (one report), STAFF.
AccessLog	Every opening and step	Append only: a database trigger refuses UPDATE, DELETE and TRUNCATE. Plain ids, no foreign keys, so it outlives what it describes.
CallbackTask, PatientFlag	Work for staff	A partial unique index allows one open critical call per report.
SummaryTemplate	Locked wording	Versioned per key and language, DRAFT or APPROVED, source human or AI.
AiSummary	Every AI attempt	Shown, refused or failed, with the checker's reason and the reviewer's verdict.
ImportBatch, ImportRow	Each LIS file and row	A sentence per bad row; the file's hash.

4. Main flows

4.1 LIS export to an SMS link

1. Staff upload the CSV (the export folder is simulated by upload, A4). Each row is validated; each problem becomes a sentence, for example "Phone number "900000001" is not a 10-digit Indian mobile number." Good rows import even when others are bad.
2. Rows of one order and test form a report. In one transaction: household, patient, report, results with rule flags, and, only when the report first becomes signed, a 72 hour link, a LINK_SENT log line and, for a critical result, a callback task due in 30 minutes.
3. After the transaction commits, the SMS is sent ("Lakshmi's Complete Blood Count report is ready. Open it here: ... The link works for 72 hours."). No result ever appears in an SMS. Delivery is recorded for the desk.
4. Uploading the same file again changes nothing; a later "Processing" row can never unsign a report.

4.2 Opening a link

The link page only checks the token. Opening the report needs a tap, because message apps fetch links to draw previews. The tap opens a session for that one report, capped at 2 hours and never past the link's 72 hours, and logs LINK_OPENED. Consent comes first. An expired or revoked link shows "We tied this page up for you" and offers a code.

4.3 Signing in with a code

Mobile number, then a six-digit code that lasts 10 minutes. The first time a number signs in: consent, then a two-step setup (how to read reports; who is on this number, with the lab's guesses filled in). Five wrong codes pause the number for 15 minutes; a resend waits 60 seconds; at most 5 codes a day (A19). An unknown number gets the same screens but no SMS. Codes are stored as keyed hashes and compared in constant time. Signing in opens the whole family's file, including pages whose links expired (A28).

4.4 Reading a report

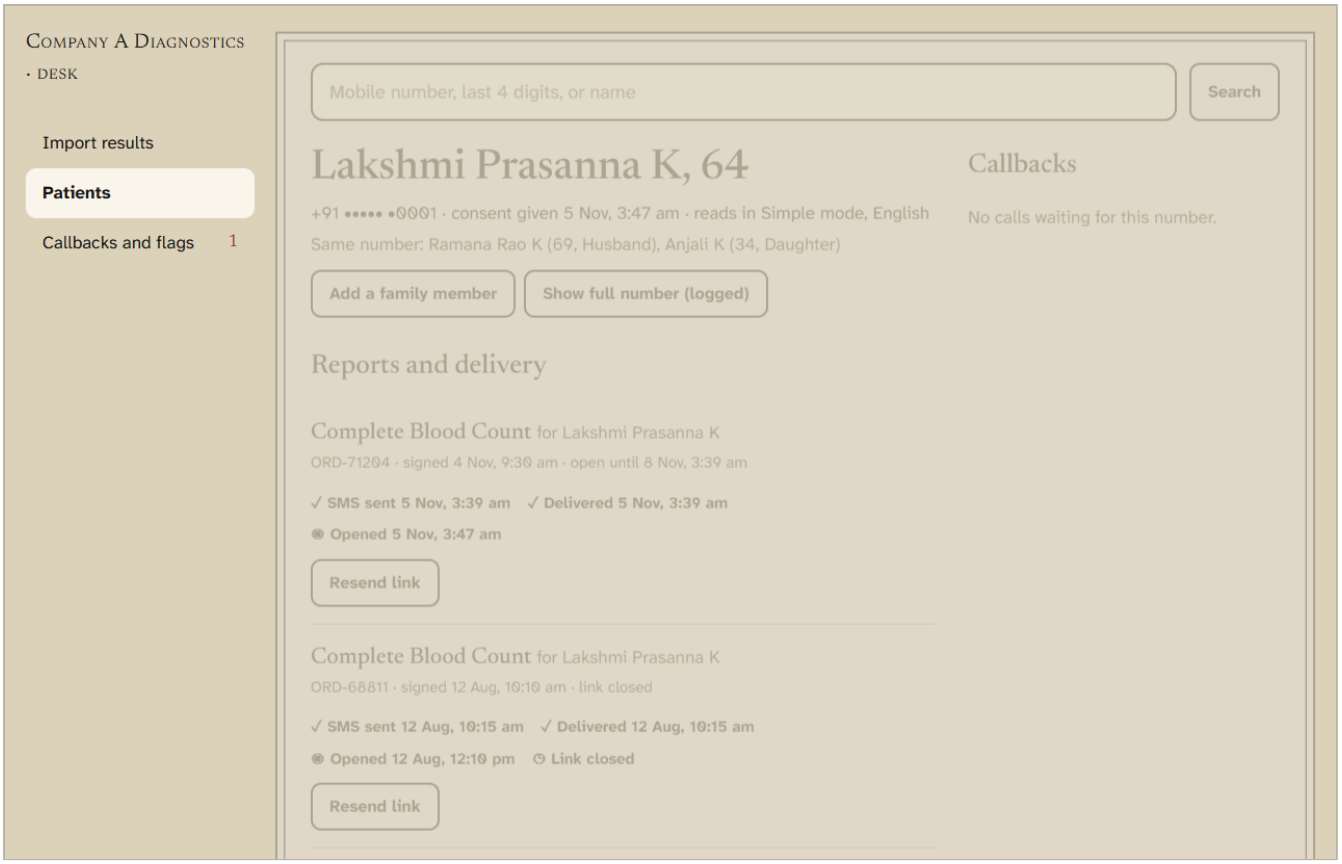
Every report read joins through the session's own household, so an ID typed into the URL can never reach another family (tested). The opening is logged in the same transaction as the link's open count. Simple mode shows one result at a time with a ruler; Compact shows all results and the trend since the last report.

4.5 Critical values

The patient sees no number, no summary and no flag words: "Please call us about your report", with the first step already ticked ("We have asked our team to call you", with the time). The callback task was created with the report's link. Staff see the value and range on the desk. The AI is never involved.

4.6 Front desk

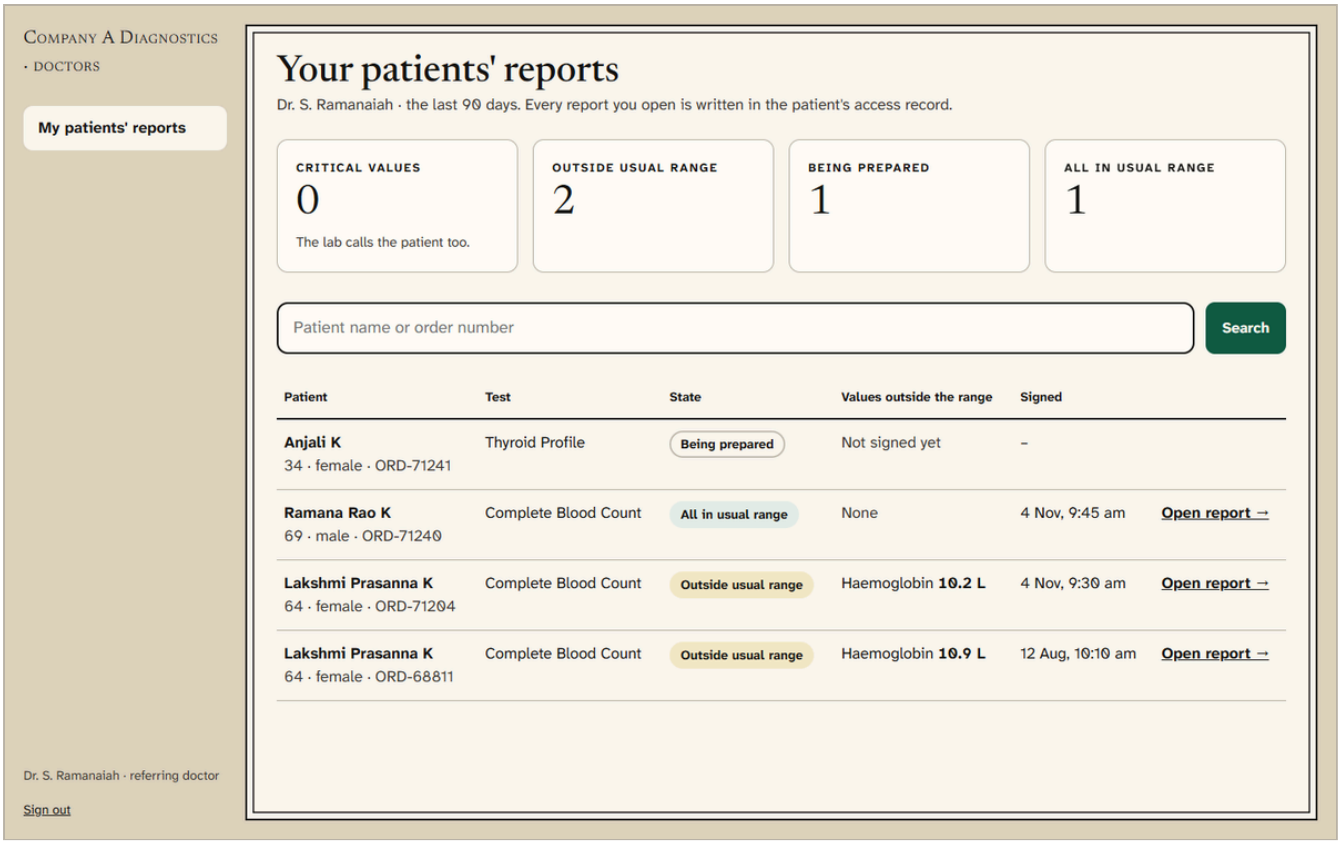
Search by full number (matched by hash), last four digits or name. Opening a record is logged; the full number is shown only on request, and that is logged too. Resend revokes the old link at once and sends a new one. Calls are marked done with a note; flags are resolved with a note.



The patient desk: the family on one number, each report with SMS, delivery and opening times and Resend, the access log, and the calls waiting for this number.

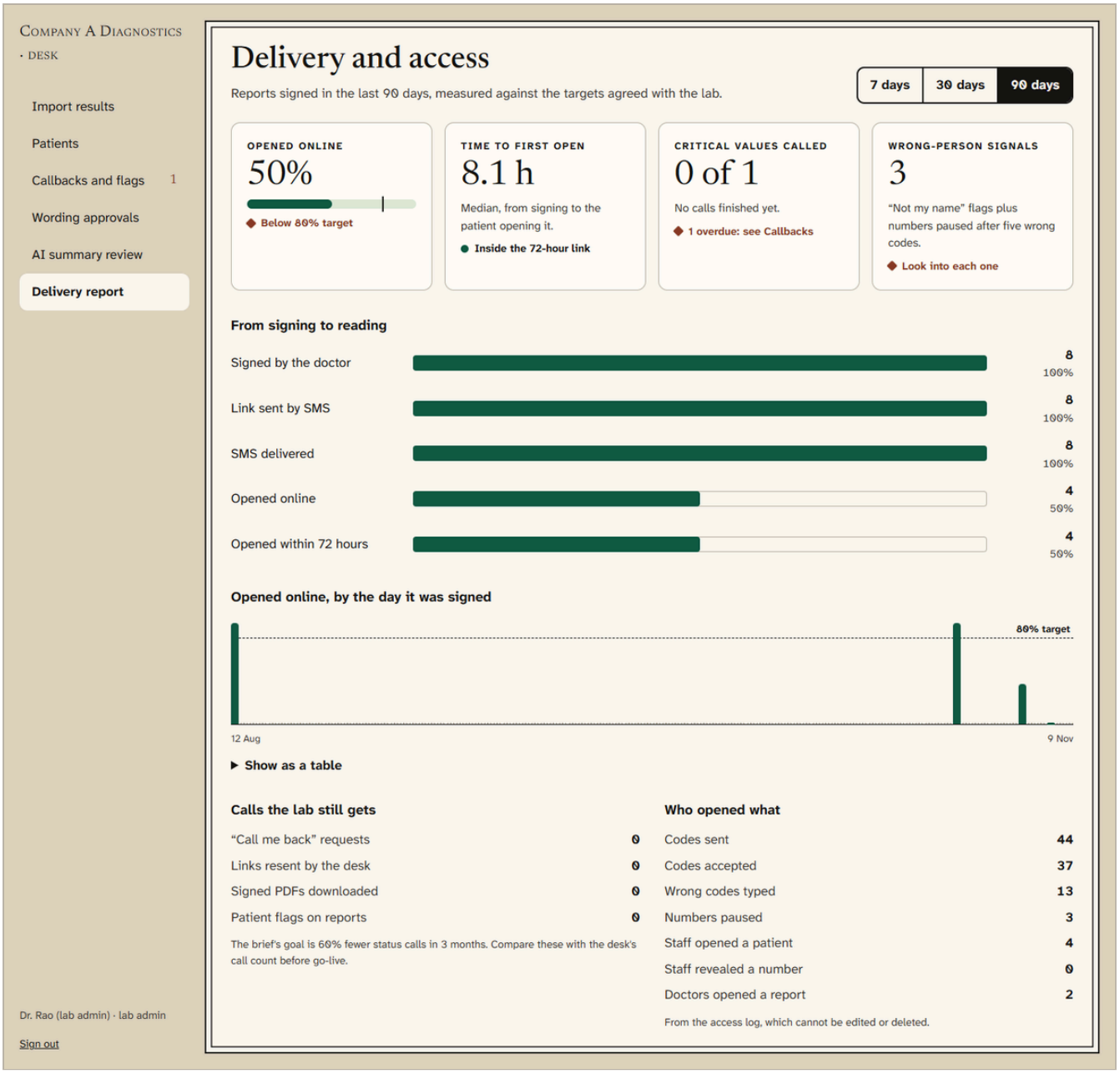
4.7 Referring doctor

Doctors sign in on the staff page with an account the lab issues. Their page lists only reports where they are the referring doctor, signed in the last 90 days or ordered then and still being prepared: critical first, then being prepared, then newest, with the out-of-range values written in each row so most reports need no click. A doctor sees full values, critical ones included, because they are the clinician, and sees whether the lab has reached a critical patient yet, so they do not call blind. They never see a phone number. Another doctor's report, an old one or an unsigned one all give the same "not found", and nothing is logged; every report actually opened is logged as DOCTOR_VIEWED_REPORT. Any desk page sends a doctor back to their own list (tested).



4.8 Delivery and access report

For the operations head (the lab admin account stands in, A31). It reads the brief's success measures from data the app already keeps. "Delivered digitally" means the patient opened the report online, not only that the SMS arrived (A33). Each headline number carries a met or not-met mark and a word. The trend chart shows one series in one colour, the 80% target as a labelled line, a tooltip per day and a table view. At 2,000 reports a day the same numbers would come from a nightly roll-up table instead of live rows.



5. How the AI part works, and what happens when it is wrong

Rules decide every flag. Locked templates supply every sentence. The AI writes only the short "In short" paragraph, and only for tests without locked wording, and only if a checker passes it.

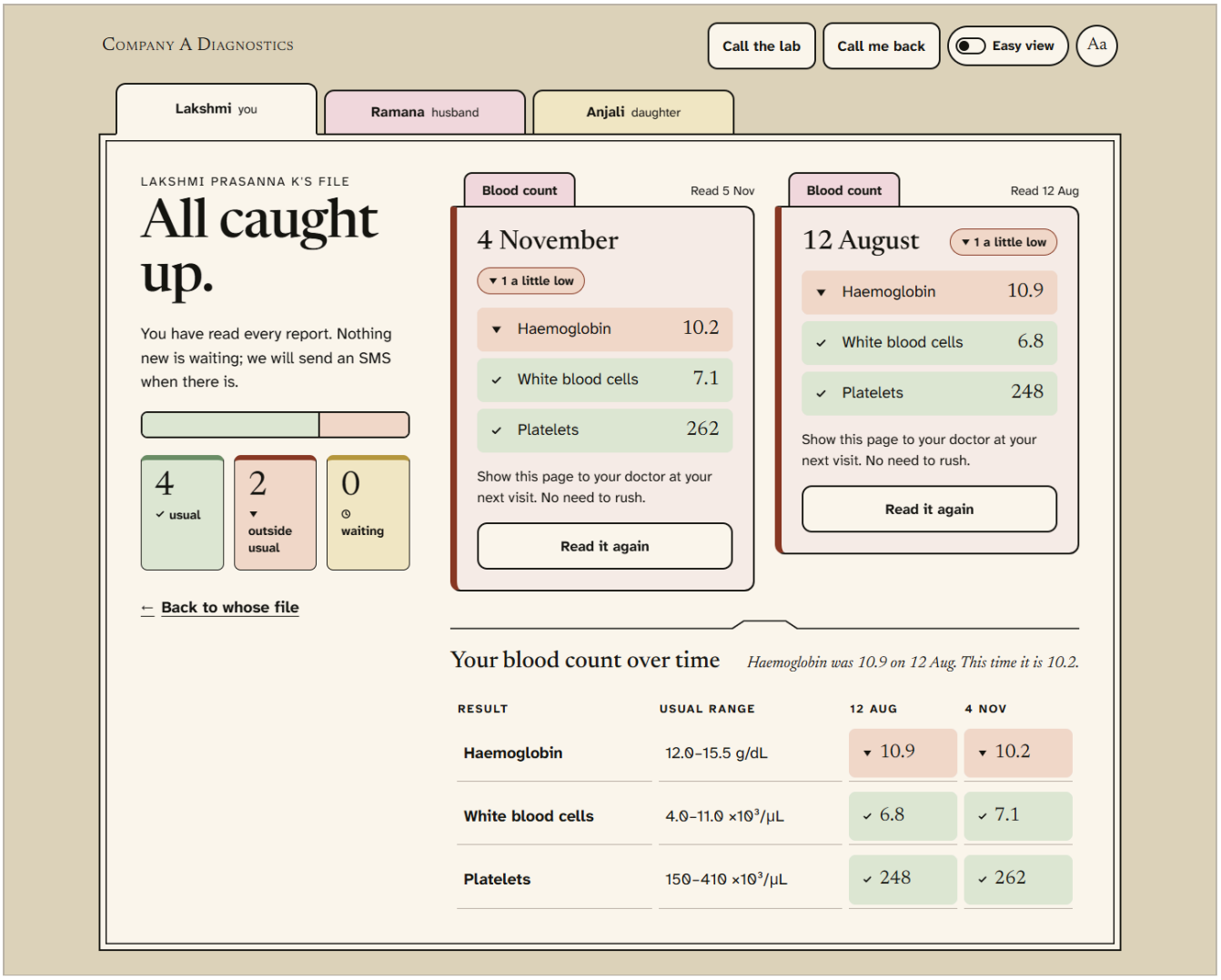
1. **Rules** set LOW, USUAL or HIGH against the range printed on the report, and "mild" or "marked" by how far past the limit (more than 20% is marked: 10.2 against 12.0 is "a little low").
2. **Templates** turn each finding into words. A test fails if any sentence the builder produces does not match an approved template.
3. **Critical reports** get no summary and no AI call.
4. **Locked tests** (the brief's top 30; CBC and Electrolytes in the demo) use templates only. The demo deliberately leaves Lipid and Thyroid unlocked so the AI path has real work to do; a go-live loads the lab's 30 tests and their approved wording.
5. **Other tests** (Lipid, Thyroid in the demo) get an AI paragraph. The model sits behind one interface: on the live demo it is Google Gemini `gemini-3.5-flash-lite` on the free tier (A34), and Claude `claude-`

`opus-5-5` at low effort is supported by setting its key instead. The model receives only test name, analyte names, values, units, printed ranges and the rule's words. No name, phone, age or order ID.

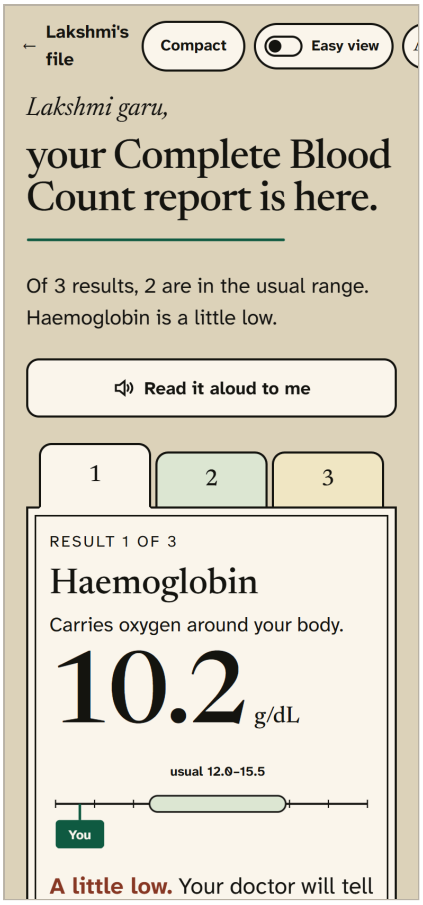
6. **The checker** refuses any number not on the report, disease or organ names, medicine or treatment advice, alarming or falsely calming words, a flag that disagrees with the rule, or a flagged result left out.
7. **When the AI is wrong or unsure:** a refused answer, an error, or no answer within 8 seconds means the patient sees the template sentence instead, at once. Every attempt is stored with the reason. With no API key set, every patient simply gets the template wording; nothing breaks.
8. **Human review:** the lab admin's review page shows unreviewed summaries next to the rule findings, counting toward the brief's 50 a week. Marking one "wrong" stops it showing immediately. Patients can also press "Something looks wrong".
9. **Translations:** the AI can draft Telugu or Hindi wording for any approved English template. Drafts keep every placeholder or are dropped, and stay hidden until a lab admin approves them; until then the patient sees English.

6. UI and design decisions

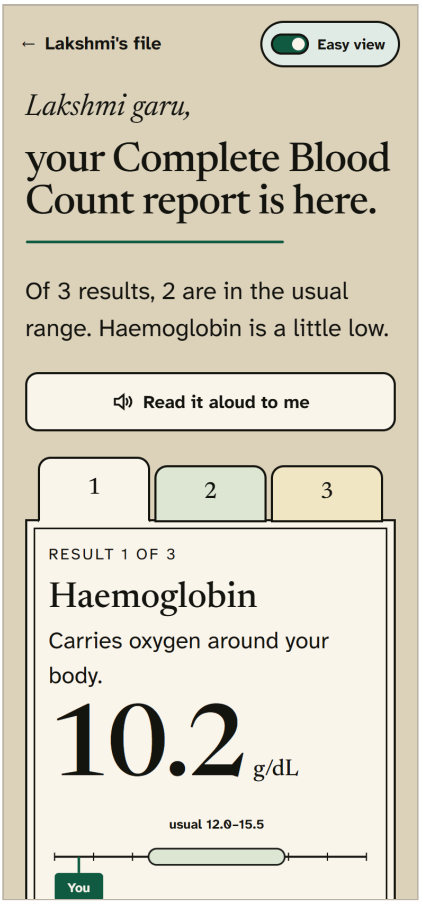
The design idea is "your health file, kept by people who know you": every report is a page in a file, behind an index tab. The client values minimal, purposeful design, and many patients are elderly on small phones, so every screen has one main section and one green next step, large type, and plain, calm words. Status is always a word plus a mark, never colour alone, so it also reads in black and white and for colour-blind readers. There is no red anywhere a patient looks. Spacing sits on a 4 px grid; all lines are clean vectors.



A person's file on a laptop: where things stand on the left (a bar of results in and outside the usual range, three counts), each page as a card with its results in the lab's rule words, and each test over time. Every sentence comes from approved templates.



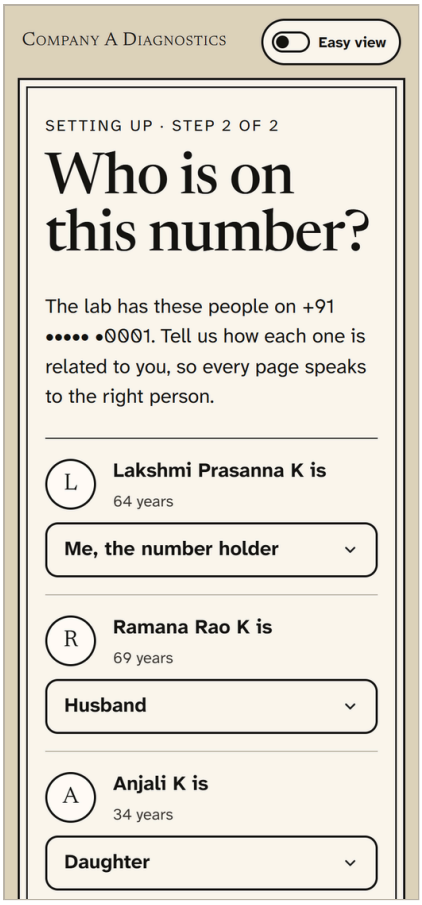
Report on a phone: one result at a time; the ruler's position comes only from the printed range.



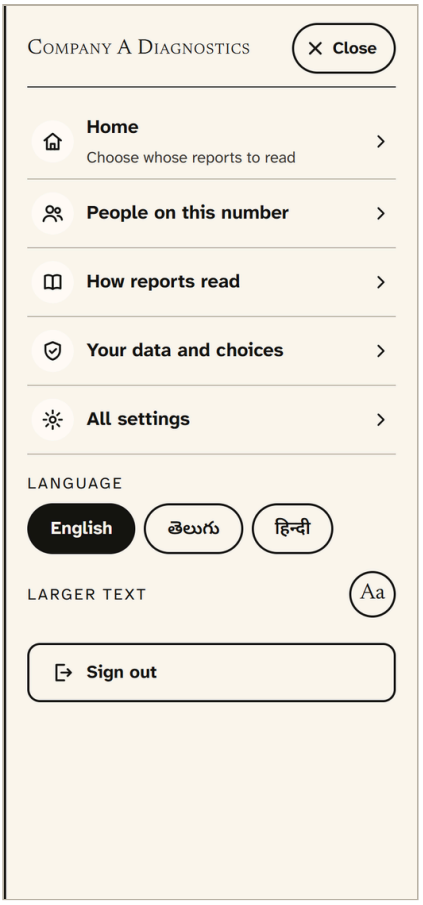
Easy view for older readers: bigger text, the Simple report only, secondary buttons step aside.



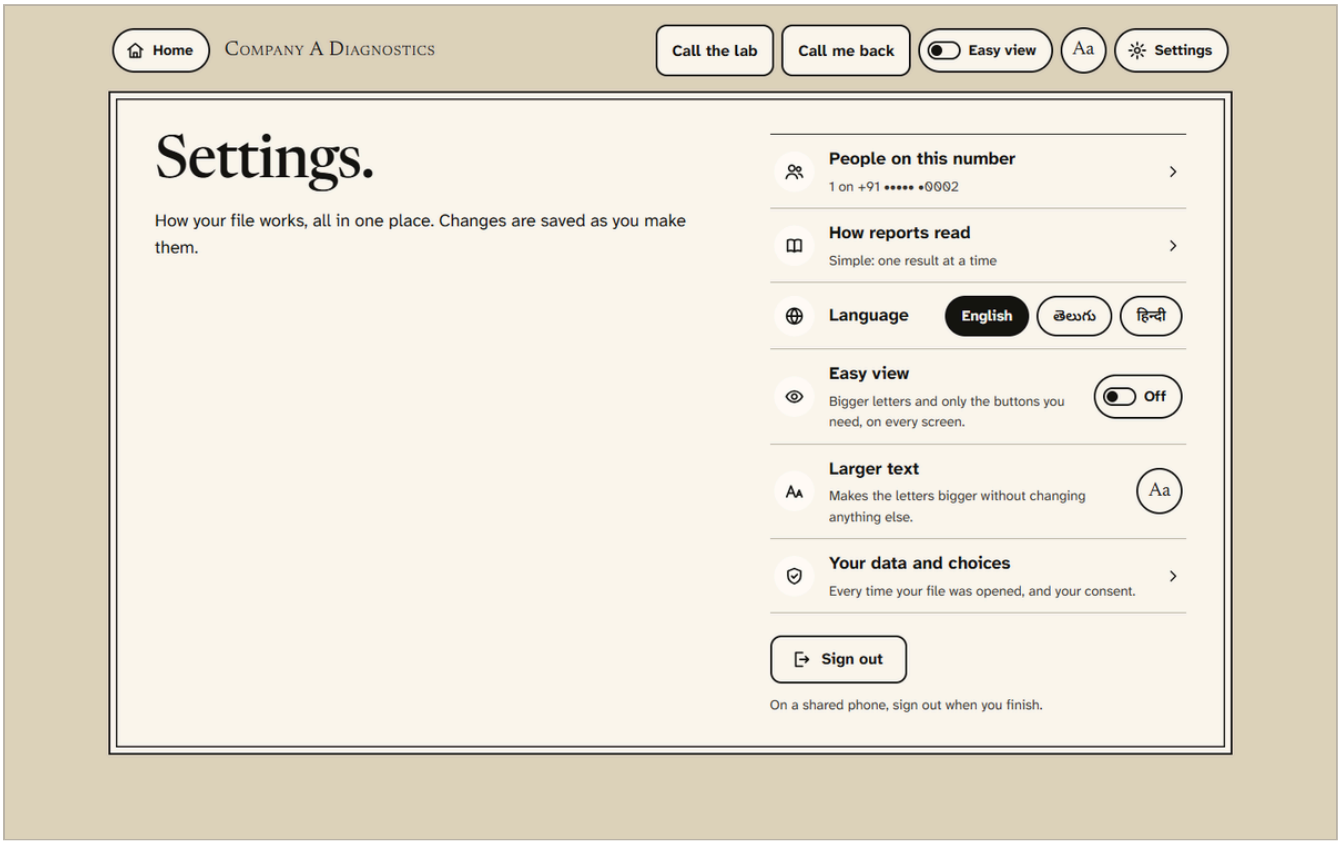
Please call us: no value, no summary, the call already queued, a fixed safety note.



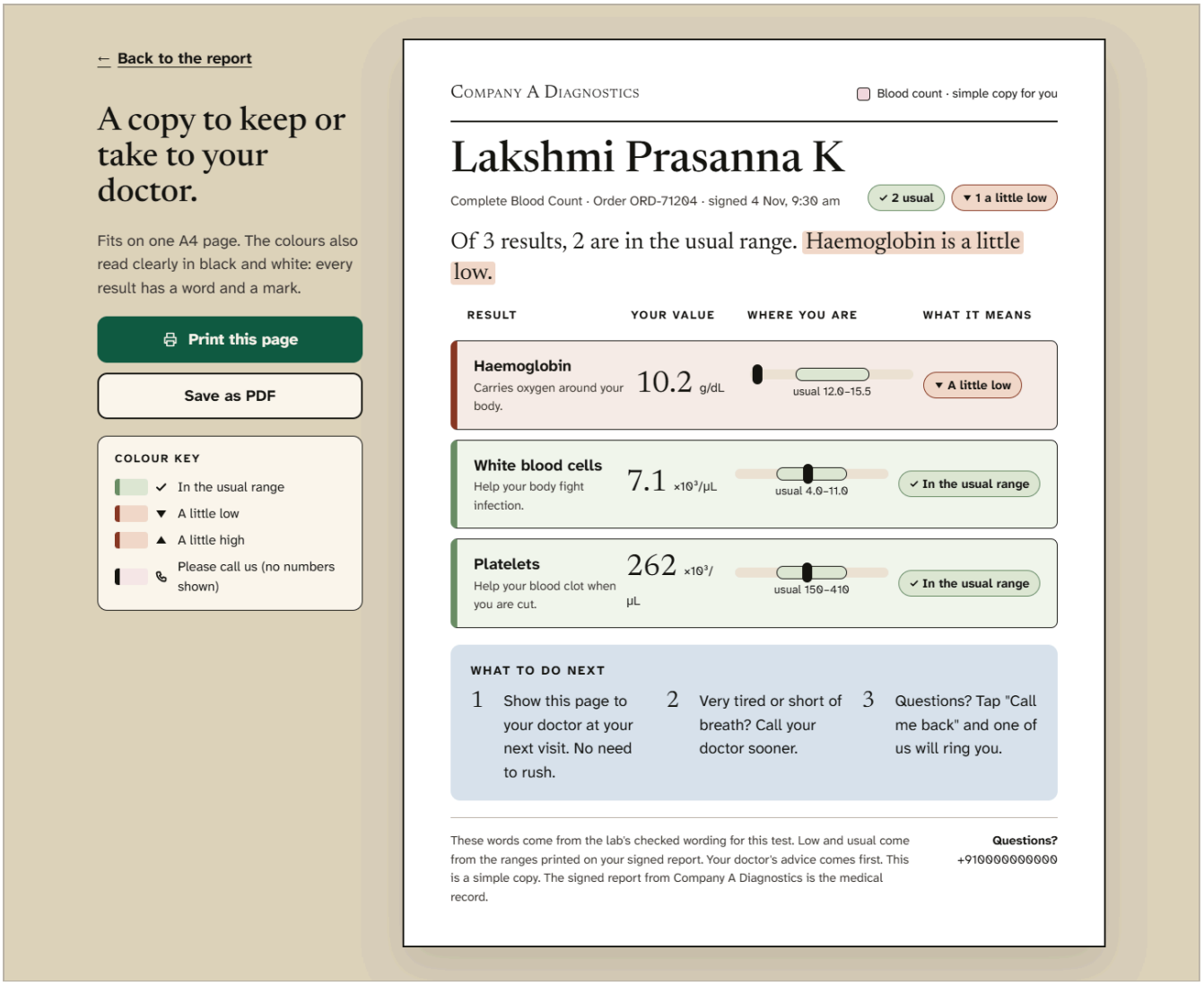
First-time setup, step 2 of 2: the lab's guesses filled in, one tap if right. Adding or removing people stays with the desk.



The phone menu: Home, people, reading style, data, language, text size, sign out. A native dialog that slides in from the side its button is on.



Settings on a laptop: one list; rows that open a page end in a chevron, rows that change at once hold their own switch. On laptops the header has Home, the call buttons, Easy view, Aa and Settings.



Printable copy: one A4 page with a colour key; Print opens the dialog, Save as PDF downloads the same simple copy. On phones the page becomes cards with a sticky Print and Save bar.

Decision	Why
One layout system: 480 px phones, 720 px tablets, up to 1280 px laptops; two columns on laptops	Families read on phones and laptops; staff on desktops. No long scrolling on a laptop, one column on a phone. Each screen's main action is checked to sit in the first screenful at 1366x768 down to 360x640.
Easy view switch on every screen, also set by choosing Simple	Older readers get bigger text and only the essentials: one clear action plus Call the lab. Pure CSS, so it switches instantly with no reload.
Home in words on every signed-in screen; the lab name is plain text	A logo that secretly links home is a convention many older users do not know. Home always means one place: choosing whose reports to read.
A menu on phones and tablets, a Settings link on laptops	Phones fit three header controls (Home, Easy view, Menu); only secondary items live in the menu, never a screen's main action. Laptops have room, so nothing hides there. Pages with a back link keep Home in the menu on phones.

Decision	Why
First-time setup in two labelled steps	People see when setup ends. Confirming who is on the number costs one tap and prevents a health report greeting the wrong person.
Desk and doctor pages: numbers lead, words explain them, met or not met is a mark plus a word	The operations head should read the page in ten seconds; nothing relies on colour alone.
Call the lab and Call me back on every screen (header on laptops, bottom bar on phones)	The brief's goal is that patients never have to hunt for help.
Tabs, Back and Next are plain links; forms are plain posts	Everything works before JavaScript loads on a slow phone.
Fonts self-hosted and split by script	No call to Google Fonts; a phone downloads Telugu glyphs only when a page shows Telugu.
Telugu and Hindi get taller line height and a smaller display size	The scripts are taller; buttons grow and never clip.
Motion only where it means something: pages settle in, a loading bar along the top, buttons press, the green arrow nudges forward, cards lift when they open, result rows arrive in order, the proportion bar grows, the ruler finds the value, the call button breathes	Each one answers a question (did my tap work, where does this go, what is this bar). Transform and opacity only; all of it stops under reduced motion.
One notice layout for expired, invalid, timeout, empty, error, withdrawn	Every dead end says what happened and what to do next, signed by the front desk. No dead buttons anywhere.

7. Security and privacy (DPDP Act 2023)

- **Consent** once, before any report, stored with its text version and time; withdrawing ends every session on the number, stops links and SMS, and is logged. A critical result still creates a staff call.
- **Access log** for every code sent, sign-in, link and report opening, PDF download, consent change, relation change, staff view, full-number view and doctor view, written in the same transaction as the action. The database refuses to change or delete it. Patients read their own record in plain words.
- **Secrets** only in environment variables; `.gitignore` was the first commit. Phone numbers hashed and encrypted; tokens and codes stored only as keyed hashes.
- **Access control** on the server for every page and action: reports through the session's household (IDOR tests), link sessions scoped to one report, staff and doctor roles per route (tested so no desk route lacks a permission, and each role is checked against every route); doctor queries scoped to their own referring-doctor id.
- **Abuse limits:** OTP lockout, cooldown and daily cap; staff lockout after 5 wrong passwords; same answers for unknown numbers and emails.
- **Inputs** validated with Zod at every boundary; errors are sentences.
- **Headers:** Referrer-Policy no-referrer (links carry tokens), framing refused, nosniff, HSTS in production, private pages never cached.

8. Testing

Kind	Count	What it proves
Unit (Vitest)	107	Flags and boundaries, critical limits, 72 hour links, phone rules, OTP limits, CSV reading and row errors, summary builder (no free text), AI checker, staff and doctor permissions.
Database (Vitest, real Postgres)	71	Idempotent import with the brief's three orders, links on sign-off, critical calls, OTP sign-in and lockout, IDOR and consent rules, link sessions, call-backs and flags, resend, staff lockout, AI summary paths with fake writers, append-only log, relation rules, doctor scoping, delivery report counts.
End to end (Playwright)	6	Sign in to a report explained by rule; a critical page with no value; five wrong codes pause the number; the menu to settings to change a relation and back Home; a staff import listing bad rows; a doctor seeing only their own patients and refused from the desk.

Logic was written test first, visibly in the git history (a `test:` commit, then a `feat:` commit).

9. Performance and accessibility

First load measured on the production build at 390 px: about 155 KB transferred per page without fonts, of which about 135 KB is the React and Next.js runtime and about 20 KB is this app's code and CSS. This misses the 150 KB budget by about 5 KB; the next step is to drop the remaining client components from pages that do not need them. Fonts are a one-time, cached download. Every screen has a loading, empty and error state. Accessibility: real buttons, links and labels; each code box labelled; the ruler is one image with a full-sentence label; 3 px green focus ring; body text contrast above 7:1; touch targets at least 48 px.

10. Assumptions

Everything the brief does not say and the build had to decide, as kept in `docs/ASSUMPTIONS.md`:

Client and scope

- A1. The client is the lab chain in Vijayawada described in the brief. An earlier build plan said Navi Mumbai; that was an error and the brief wins.
- A2. Roles built in the core slice are patient and front-desk staff, as the brief asks. A `lab_admin` role is added only to approve wording (templates and translations), because front-desk staff should not change medical wording.
- A3. Referring doctor view and the operations delivery report are optional depth. They are built only if the core slice is finished.

Lab system and delivery

- A4. The LIS export folder is simulated by uploading its CSV in the staff console. The import service takes a file, so a folder watcher can call the same code later. No change to the LIS, which is out of scope.
- A5. A report counts as signed when its export row says `Ready` or `Ready (critical)`. `Processing` means tested but not signed, and is never shown to the patient as a result.
- A6. Messages go out as SMS through a mock provider. WhatsApp, which the brief also allows, would be a second implementation of the same interface.
- A7. The signed PDF comes from the LIS and is not in the sample data, so the demo serves a generated stand-in clearly marked as a sample.

Links and sign-in

- A8. Read literally, the brief says a link opens its one report for 72 hours, and after that the patient signs in with an OTP to reopen it. The build follows that reading.

Data and hosting

- A9. "Hosting in India" is met with the database in Supabase's Mumbai region and Vercel functions pinned to the Mumbai region.
- A10. Seed data uses the brief's sample patients, orders and results. Phone numbers are fake (+91 90000 0000x), and extra family members are invented.

AI summaries

- A11. The AI makes real model calls; it is a required feature, not a mock. Rules decide every flag. The AI only writes the wording, and a checker rejects any output whose numbers or flag words do not match the rules, or that uses banned words. On a rejected output, a timeout or no API key, the patient sees the plain rule line and the full report link.
- A12. The AI receives only the test name, analyte names, values, units, ranges and rule flags. No name, phone, age or order ID is sent, so no personal data leaves India. An India-hosted model (for example Sarvam) can replace it behind the same interface.
- A13. Critical results never reach the AI. Tests with an approved locked template use the template text and make no AI call at read time.
- A14. Telugu and Hindi wording written by the AI is a draft until a lab_admin approves it. Until then the patient sees English.

- A22. The AI model is claude-opus-5-5 at low effort, set by AI_MODEL. It gets one try with an 8 second limit and no retries; after a failed or refused attempt it is not asked again for that report for 24 hours, and the patient sees the rule-based wording.

Import format and test catalog

- A15. The LIS export has the brief's sample columns (Order ID, Patient, Parameter, Result, Reference Range, Status, Referring Doctor) plus Phone, Age, Sex, Test Code, and optional Relation and Signed At. The brief does not give the full layout, and a patient cannot be reached without a phone number.
- A16. A patient is matched by phone number plus name. A new name on a known number becomes a new family member on that number.
- A17. Critical limits (for example potassium 6.0 mmol/L and above, haemoglobin 7.0 g/dL and below) are placeholders. The brief gives none, so the lab doctor must confirm them before go-live. A report is also critical whenever the LIS marks it "Ready (critical)".
- A18. CBC and Electrolytes have locked, doctor-approved templates in the demo; Lipid and Thyroid do not, so the demo shows both the template path and the AI path.

Sign-in and sessions

- A19. The brief asks for a daily cap on codes but gives no number. The cap is 5 codes per number in 24 hours.
- A20. A patient session ends after 30 minutes without use and after 12 hours at most. A session opened from an SMS link lasts at most 2 hours and opens only that report. Staff sessions end after 2 hours idle or 10 hours.
- A21. Three demo numbers (+91 90000 00001 to 00003) accept a fixed code only when DEMO_MODE is on, so reviewers can sign in without a phone. It is off by default.
- A23. In demo mode only, the three demo numbers skip the 60 second resend wait and the daily cap, because their code is fixed and no SMS is sent; many reviewers may sign in on the same day. The five-wrong-tries pause still applies.

Staff work and patient requests

- A24. The staff desk is in English only. Telugu and Hindi are for patients.
- A25. The brief gives no call times. A critical result asks staff to call within 30 minutes; "Call me back" within 2 hours, between 8 am and 8 pm.
- A26. "Call me back" needs a sign-in or an open link, so the lab knows whose file it is. Signed out, the page offers the lab's phone number instead of promising a call.
- A27. "Something looks wrong" creates a flag for staff. Nothing on the report changes until a doctor has looked.
- A28. Signing in with a code opens every signed report on that number, including ones whose SMS link has expired. The code is the "OTP to reopen" the brief asks for.
- A29. People on a number come only from the lab's records (the LIS export or the front desk). From the phone, the number holder can say how each person is related to them, from a fixed list, but cannot add or remove anyone, so no one's reports can be hidden or claimed by mistake. A relation the lab typed that is not on the list (for example "Daughter-in-law") is kept as it is. Only one person can be "Self".
- A30. The first time a number signs in, after consent and reading style, the number holder confirms who is on the number. It is one screen with the lab's guesses already filled in, so it can be passed with one tap.

- A31. The brief names an operations head but no separate account for one. The lab admin account opens the delivery and access report; a real deployment would add an operations role with only that page.
- A32. Referring doctors sign in with an email and password the lab issues, on the same sign-in page as staff, and see only reports where they are the referring doctor: signed in the last 90 days, or ordered in the last 90 days and still being prepared. They see full values, critical ones included, because they are the treating clinician. They never see a patient's phone number. Every report they open is logged.
- A33. "Delivered digitally" in the operations report means the patient opened the report online at least once (by link or by code). A delivered SMS alone does not count, because the brief's goal is that the patient reads it without calling or visiting.
- A34. The live demo writes the AI paragraph with Google Gemini (`gemini-3.5-flash-lite`) on Google's free tier, to keep the demo at no cost. On that tier Google may use prompts to improve its products; this is acceptable only because the app sends de-identified values (test, analyte names, values, units, printed ranges, rule words), never a name, phone, age or order number. A real deployment would use a paid tier (prompts not used for training) or an India-hosted model behind the same interface. Claude remains supported by setting its key.

11. What was cut, honest limits, and what comes next

- **Built beyond the core:** the referring doctor view and the delivery and access report (both optional depth in the brief), patient settings, and first-time setup.
- **Cut:** a separate operations-head role (the lab admin opens the report, A31); WhatsApp (SMS only, behind the same interface); doctors cannot add notes or message the lab.
- **Mocked by design:** SMS (provider interface ready for an SMS or WhatsApp gateway) and the signed PDF (a marked stand-in).
- **Limits:** the demo has 4 tests, 2 with locked wording, not the brief's 30; Telugu and Hindi copy comes from the design and needs native review; critical limits and the safety note are placeholders for the lab doctor; correcting a phone number on file is not built; there is no folder watcher yet (the import service is ready for one); the page-weight budget is missed by about 5 KB.
- **Next:** a real SMS or WhatsApp gateway with delivery receipts; the lab's 30 most common tests with approved wording; a nightly roll-up for the delivery report at full volume; an operations-head role; a paid AI tier or an India-hosted model behind the same interface; native-speaker review and doctor sign-off of all wording; a folder watcher for the LIS export; a usability test of easy view with five patients over 60.